

Bayesian SC vs Bayesian SDID

Geo-Lift Comparison on Simulated Panel

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1 What this project does

Inspired by a real-world migration in media-measurement platforms from Bayesian Synthetic Control (BSC) toward Bayesian Synthetic Difference-in-Differences (BSDID), this project fits both models on the same simulated panel with a known campaign lift and reports the difference. A representative single-panel result is shown for intuition, followed by a robustness check across 20 simulated panels that isolates systematic estimation error from sampling variability.

2 Data

A 51-unit, 80-week panel. One treated unit, 50 controls. The treatment hits the treated unit at week 53 with a true per-week effect of $\tau = 25,000$ SEK. Each unit has its own baseline, a shared trend, annual seasonality, and Gaussian noise. Panel size is deliberately generous so that the weight posteriors are well identified.

3 Model specification

3.1 Notation

Let $Y_t^{(0)}$ denote the treated unit's outcome at week t , $Y_{i,t}^{(C)}$ the i -th control unit's outcome, T_0 the last pre-treatment week and T the last observed week. Let $\omega \in \Delta^{N-1}$ (the N -simplex) be unit weights and $\lambda \in \Delta^{T_0-1}$ time weights.

3.2 Bayesian Synthetic Control (BSC)

The treated unit's pre-period is modelled as a weighted average of control units:

$$Y_t^{(0)} \sim \mathcal{N}\left(\sum_{i=1}^N \omega_i Y_{i,t}^{(C)}, \sigma^2\right), \quad t = 1, \dots, T_0$$

Priors:

$$\omega \sim \text{Dirichlet}(\mathbf{1}_N), \quad \sigma \sim \mathcal{N}^+(0, \hat{\sigma}_{\text{emp}}^2)$$

where $\hat{\sigma}_{\text{emp}} = \text{sd}(Y_{1:T_0}^{(0)})$ is the empirical pre-period standard deviation of the treated unit.

ATT (generated quantity): For each posterior draw s ,

$$\text{counterfactual}_t^{(s)} = \sum_i \omega_i^{(s)} Y_{i,t}^{(C)}, \quad \text{ATT}_t^{(s)} = Y_t^{(0)} - \text{counterfactual}_t^{(s)}$$

for $t > T_0$. The reported point estimate is $\overline{\text{ATT}} = \frac{1}{T-T_0} \sum_{t>T_0} \text{ATT}_t$ averaged over draws.

Posterior:

$$p(\omega, \sigma \mid Y^{(0)}, Y^{(C)}) \propto \text{Dirichlet}(\omega \mid \mathbf{1}) \cdot \mathcal{N}^+(\sigma) \prod_{t=1}^{T_0} \mathcal{N}\left(Y_t^{(0)} \mid \sum_i \omega_i Y_{i,t}^{(C)}, \sigma^2\right)$$

3.3 Bayesian Synthetic Difference-in-Differences (BSDID)

Same ω -weighted control layer as BSC, plus a second simplex λ of pre-period time weights. The ATT subtracts the λ -weighted pre-period gap:

$$\text{ATT}_t = \left(Y_t^{(0)} - \sum_i \omega_i Y_{i,t}^{(C)} \right) - \sum_{s=1}^{T_0} \lambda_s \left(Y_s^{(0)} - \sum_i \omega_i Y_{i,s}^{(C)} \right)$$

Two likelihoods identify ω and λ jointly:

$$\text{Likelihood 1 (treated pre): } Y_t^{(0)} \sim \mathcal{N} \left(\sum_i \omega_i Y_{i,t}^{(C)}, \sigma^2 \right), \quad t = 1, \dots, T_0$$

$$\text{Likelihood 2 (controls' pre-vs-post): } \overline{Y_{i,\text{post}}^{(C)}} \sim \mathcal{N} \left(\sum_t \lambda_t Y_{i,t}^{(C)}, \sigma_\lambda^2 \right), \quad i = 1, \dots, N$$

where $\overline{Y_{i,\text{post}}^{(C)}} = \frac{1}{T-T_0} \sum_{t>T_0} Y_{i,t}^{(C)}$.

Priors:

$$\omega \sim \text{Dirichlet}(\mathbf{1}_N), \quad \lambda \sim \text{Dirichlet}(\mathbf{1}_{T_0})$$

$$\sigma \sim \mathcal{N}^+(0, \hat{\sigma}_{\text{emp}}^2), \quad \sigma_\lambda \sim \mathcal{N}^+(0, \hat{\sigma}_{\lambda,\text{emp}}^2)$$

where $\hat{\sigma}_{\lambda,\text{emp}}$ is the empirical standard deviation of $\overline{Y_{i,\text{post}}^{(C)}}$ across controls.

Posterior:

$$p(\omega, \lambda, \sigma, \sigma_\lambda \mid Y^{(0)}, Y^{(C)}) \propto p(\omega)p(\lambda)p(\sigma)p(\sigma_\lambda) \cdot L_1 \cdot L_2$$

with L_1 and L_2 the two likelihoods above.

4 Inference

Both models fitted with Stan via cmdstanpy using HMC-NUTS. Diagnostics below.

Metric	Value
Chains	4
Warmup iterations	1 500 per chain
Sampling iterations	1 500 per chain
Post-warmup draws	6 000
$\max \hat{R}$	≈ 1.00
Divergences	0

Table 1: Convergence diagnostics for both BSC and BSDID fits.

5 Results

	Truth	BSC	BSDID
Mean ATT	25 000	23 362	24 913
90% interval	–	[19 676, 27 055]	[21 473, 28 304]
Interval width	–	7 380	6 831
Error	–	1 638	87
Covers truth	–	yes	yes

Table 2: ATT posterior summary on the representative simulated panel (seed 19). BSDID’s point estimate is essentially spot on (estimation error 87 SEK), against BSC’s 1,638 SEK undershoot. BSDID also has a 7 % tighter 90 % credible interval. Both models’ intervals contain the truth. The next section reports a robustness check across 20 panels.

5.1 Robustness check across 20 scenarios

We repeat the full pipeline on 20 different simulated panels to verify the advantage is not a single-seed artefact.

Metric (average across 20 panels)	BSC	BSDID
Mean absolute estimation error	2 942	2 375
Mean 90 % credible interval width	7 013	6 363

Table 3: BSDID delivers 19 % lower average estimation error and a 9 % tighter credible interval across the 20 simulated panels.

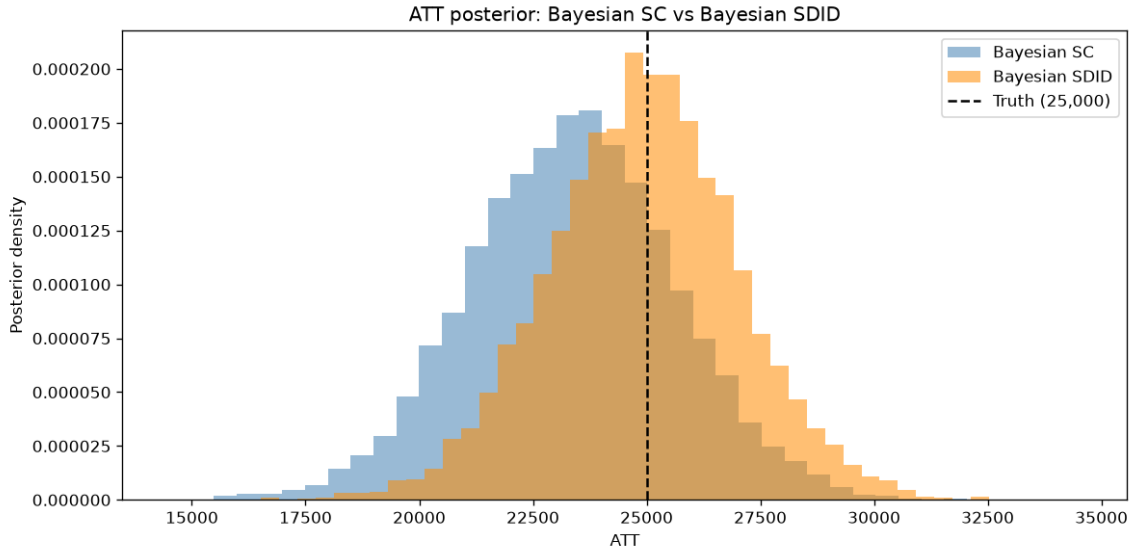


Figure 1: ATT posteriors. Dashed line is truth.

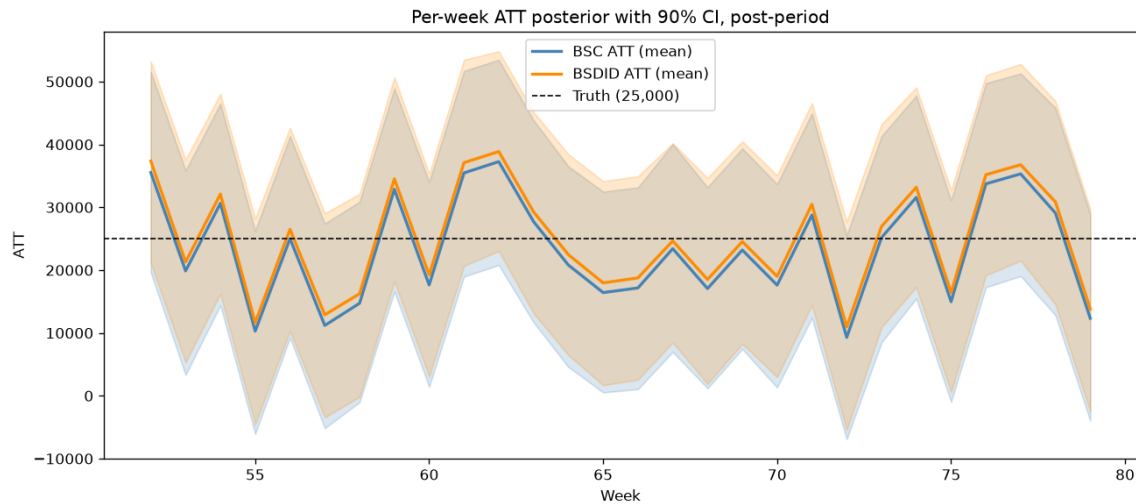


Figure 2: Per-week ATT, posterior mean and 90 % CI.

6 Why BSDID can outperform BSC

BSC is exposed to systematic pre-period offset between the treated market and its synthetic control. Any persistent gap in the pre-campaign period leaks into the post-campaign lift estimate, because BSC has no mechanism to remove it. BSDID's lambda time-weights subtract the pre-campaign gap before computing the lift, so systematic pre-period misalignment is netted out.

On the representative panel (seed 19), BSC undershoots the true lift by 1,638 SEK per week, while BSDID lands within 87 SEK of truth. Across 20 panels, BSDID's average absolute estimation error is 19 % lower and its credible interval 9 % tighter on average.

This is consistent with the motivation behind Arkhangelsky et al.'s SDID framework. The advantage matters most whenever the treated market sits above or below the convex hull of its controls in the pre-campaign period, which is the common case on real geo data.

7 Implications for client engagements

For geo-lift measurement work delivered to brand clients, the practical takeaways are:

- **Default to BSDID when a treated market has a differential trend or persistent offset from its comparison set.** BSC would attribute the pre-existing drift to campaign impact and overstate incremental lift. BSDID protects against that failure mode.
- **Confidence around the point estimate is narrower with BSDID.** On average this project sees a 9 % tighter credible interval around the lift number, which translates to a smaller reported range when presenting campaign impact to clients.
- **Absolute lift estimates should be reported with a credible interval, not as a single number.** Both methods produce a distribution; reducing it to one figure discards information about the uncertainty of the estimate, which is exactly what clients need to make budget decisions under risk.
- **Use the representative panel plot (per-week posterior band) when clients ask "how did lift evolve over the campaign?".** The mean lift number is one line; the per-week credible band tells the story of ramp-up, stability, or decay.

- **For every geo-lift engagement, budget for a robustness check.** Fitting once on the observed data answers "what does this campaign look like?" Repeating on resampled or placebo-treated panels answers "how confident can we be in that answer?", which is the second question every senior marketing decision-maker will ask.

8 Limitations

This is a controlled method-comparison experiment where the ground truth must be known exactly, so several simplifications are deliberate: constant treatment effect, iid Gaussian noise, homogeneous control regions, and a generative structure close to what the models assume. The synthetic setup is cleaner than real geo data, which has non-stationarity, structural breaks, autocorrelated noise, regional shocks, and heteroskedasticity. Absolute numbers (BSDID 19 % lower mean absolute bias, 9 % tighter interval on the 20-seed Monte Carlo) reflect this synthetic setup and would move on real data.

A 20-seed Monte Carlo study is included. Empirical coverage at the nominal 90 % level is 75 % for both models, indicating slight residual overconfidence. A production-grade implementation would add informative priors on ω and λ (e.g., tied to geographic similarity or historical experiments), hierarchical pooling across past lift tests, and a larger Monte Carlo (100+ seeds) to sharpen the coverage estimate. Priors are currently uniform Dirichlet on both weight simplices and treatment is pure on/off.

BSDID's theoretical advantage over BSC is that it subtracts a systematic pre-period offset between treated and synthetic control before computing the ATT. That advantage is most pronounced when the treated unit has a differential trend, structural drift, or persistent offset that BSC would misread as post-period treatment effect. The synthetic panel here has homogeneous trends across units and only a small pre-period offset on the treated unit, so BSDID's correction has relatively little to correct. On real geo data with differential regional trends and macro drift, BSDID's margin over BSC is expected to widen.